

Air University



Computer Networks (LAB)

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Lab Task

Router A:

```
Router-A#show ip int brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 unassigned      YES unset   administratively down down
FastEthernet0/1 unassigned      YES unset   administratively down down
Serial0/3/0    unassigned      YES unset   administratively down down
Serial0/3/1    unassigned      YES unset   administratively down down
Vlan1          unassigned      YES unset   administratively down down
Router-A#
```

```
Router-A#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router-A(config)#int serial 0/3/0
Router-A(config-if)#ip address 1.0.0.1 255.0.0.0
Router-A(config-if)#encapsulation ppp
Router-A(config-if)#bandwidth 1000
Router-A(config-if)#clock rate 64000
Router-A(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
Router-A(config-if)#end
Router-A#
%SYS-5-CONFIG_I: Configured from console by console
```

```
Router-A(config)#int serial 0/3/1
Router-A(config-if)#ip address 4.0.0.2 255.0.0.0
Router-A(config-if)#
Router-A(config-if)#encapsulation ppp
Router-A(config-if)#bandwidth 1000
Router-A(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/1, changed state to down
Router-A(config-if)#
Router-A(config-if)#end
Router-A#
%SYS-5-CONFIG_I: Configured from console by console
```

```
Router-A#show ip int brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 unassigned      YES unset   administratively down down
FastEthernet0/1 unassigned      YES unset   administratively down down
Serial0/3/0    1.0.0.1         YES manual  down        down
Serial0/3/1    4.0.0.2         YES manual  down        down
Vlan1          unassigned      YES unset   administratively down down
Router-A#
```

```

Router-A#show ip protocols
Router-A#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router-A(config)#router ospf50
      ^
% Invalid input detected at '^' marker.

Router-A(config)#router ospf 50
Router-A(config-router)#network 1.0.0.0 0.255.255.255 area 10
Router-A(config-router)#network 4.0.0.0 0.255.255.255 area 10
Router-A(config-router)#end
Router-A#
%SYS-5-CONFIG_I: Configured from console by console

Router-A#show ip protocols

Routing Protocol is "ospf 50"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 4.0.0.2
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    1.0.0.0 0.255.255.255 area 10
    4.0.0.0 0.255.255.255 area 10
  Routing Information Sources:
    Gateway         Distance      Last Update
    4.0.0.2          110          00:00:09
  Distance: (default is 110)

```

Router B:

```

Router-B#show ip int brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          unassigned      YES unset    administratively down down
FastEthernet0/1          unassigned      YES unset    administratively down down
Serial0/3/0              unassigned      YES unset    administratively down down
Serial0/3/1              unassigned      YES unset    administratively down down
Vlan1                    unassigned      YES unset    administratively down down
Router-B#

Router-B#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router-B(config)#int serial 0/3/0
Router-B(config-if)#ip address 1.0.0.2 255.0.0.0
Router-B(config-if)#encapsulation ppp
Router-B(config-if)#bandwidth 1000
Router-B(config-if)#clock rate 64000
This command applies only to DCE interfaces
Router-B(config-if)#no shut down

Router-B(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

Router-B(config-if)#end
Router-B#
%SYS-5-CONFIG_I: Configured from console by console

Router-B#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router-B(config)#int serial 0/3/1
Router-B(config-if)#ip address 2.0.0.1 255.0.0.0
Router-B(config-if)#encapsulation ppp
Router-B(config-if)#bandwidth 1000
Router-B(config-if)#clock rate 64000
Router-B(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/1, changed state to down
Router-B(config-if)#end
Router-B#
%SYS-5-CONFIG_I: Configured from console by console

```

```
Router-B#show ip int brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 200.100.50.1    YES manual up          up
FastEthernet0/1 unassigned      YES unset  administratively down down
Serial0/3/0     1.0.0.2         YES manual up          up
Serial0/3/1     2.0.0.1         YES manual up          up
Vlan1           unassigned      YES unset  administratively down down
Router-B#
```

```
Router-B(config)#int f 0/0
Router-B(config-if)#ip address 200.100.50.1 255.255.255.0
Router-B(config-if)#no shut down

Router-B(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router-B(config-if)#end
Router-B#
%SYS-5-CONFIG_I: Configured from console by console
```

```
Router-B#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

```
    1.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       1.0.0.0/8 is directly connected, Serial0/3/0
C       1.0.0.1/32 is directly connected, Serial0/3/0
L       1.0.0.2/32 is directly connected, Serial0/3/0
    2.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       2.0.0.0/8 is directly connected, Serial0/3/1
L       2.0.0.1/32 is directly connected, Serial0/3/1
C       2.0.0.2/32 is directly connected, Serial0/3/1
    200.100.50.0/24 is variably subnetted, 2 subnets, 2 masks
C       200.100.50.0/24 is directly connected, FastEthernet0/0
L       200.100.50.1/32 is directly connected, FastEthernet0/0
```

```

Router-B#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router-B(config)#router ospf 50
Router-B(config-router)#network 1.0.0.0 0.255.255.255 area 10
Router-B(config-router)#networ
00:44:43: %OSPF-5-ADJCHG: Process 50, Nbr 4.0.0.2 on Serial0/3/0 from LOADING to FULL, Loading
Done

% Incomplete command.
Router-B(config-router)#network 2.0.0.0 0.255.255.255 area 10
Router-B(config-router)#end
Router-B#
%SYS-5-CONFIG_I: Configured from console by console

Router-B#show ip protocols

Routing Protocol is "ospf 50"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 200.100.50.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    1.0.0.0 0.255.255.255 area 10
    2.0.0.0 0.255.255.255 area 10
  Routing Information Sources:
    Gateway         Distance      Last Update
    4.0.0.2          110          00:00:34
    200.100.50.1     110          00:00:14
  Distance: (default is 110)

```

```

Router-B#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

```

Gateway of last resort is not set

```

      1.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       1.0.0.0/8 is directly connected, Serial0/3/0
C       1.0.0.1/32 is directly connected, Serial0/3/0
L       1.0.0.2/32 is directly connected, Serial0/3/0
      2.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       2.0.0.0/8 is directly connected, Serial0/3/1
L       2.0.0.1/32 is directly connected, Serial0/3/1
C       2.0.0.2/32 is directly connected, Serial0/3/1
O       3.0.0.0/8 [110/200] via 2.0.0.2, 00:07:34, Serial0/3/1
O       4.0.0.0/8 [110/200] via 1.0.0.1, 00:10:30, Serial0/3/0
      200.100.50.0/24 is variably subnetted, 2 subnets, 2 masks
C       200.100.50.0/24 is directly connected, FastEthernet0/0

```

Router C:

```

Router-C#show ip int brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 unassigned      YES unset    administratively down down
FastEthernet0/1 unassigned      YES unset    administratively down down
Serial0/3/0     unassigned      YES unset    administratively down down
Serial0/3/1     unassigned      YES unset    administratively down down
Vlan1          unassigned      YES unset    administratively down down
Router-C#

```

```
Router-C>en
Router-C#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router-C(config)#int serial 0/3/1
Router-C(config-if)#ip address 2.0.0.2 255.0.0.0
Router-C(config-if)#encapsulation ppp
Router-C(config-if)#bandwidth 1000
Router-C(config-if)#clock rate 64000
This command applies only to DCE interfaces
Router-C(config-if)#no shutdown

Router-C(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed state to up

Router-C(config-if)#end
Router-C#
%SYS-5-CONFIG_I: Configured from console by console

Router-C#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router-C(config)#int serial 0/3/0
Router-C(config-if)#ip address 3.0.0.1 255.0.0.0
Router-C(config-if)#encapsulation ppp
Router-C(config-if)#bandwidth 1000
Router-C(config-if)#clock rate 64000
Router-C(config-if)#no shut down

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
Router-C(config-if)#
Router-C(config-if)#end
Router-C#
%SYS-5-CONFIG_I: Configured from console by console


Router-C#show ip int brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          unassigned      YES unset   administratively down down
FastEthernet0/1          unassigned      YES unset   administratively down down
Serial0/3/0              3.0.0.1         YES manual  down        down
Serial0/3/1              2.0.0.2         YES manual  up          up
Vlan1                    unassigned      YES unset   administratively down down
Router-C#


Router-C#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    2.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       2.0.0.0/8 is directly connected, Serial0/3/1
C       2.0.0.1/32 is directly connected, Serial0/3/1
L       2.0.0.2/32 is directly connected, Serial0/3/1
    3.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       3.0.0.0/8 is directly connected, Serial0/3/0
L       3.0.0.1/32 is directly connected, Serial0/3/0
C       3.0.0.2/32 is directly connected, Serial0/3/0
```

```
Router-C#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router-C(config)#router ospf 50
Router-C(config-router)#network 2.0.0.0 0.255.255.255 area 10
Router-C(config-router)#network 3.0.0.0 0.2
00:46:51: %OSPF-5-ADJCHG: Process 50, Nbr 200.100.50.1 on Serial0/3/1 from LOADING to FULL,
Loading Done
^
% Invalid input detected at '^' marker.

Router-C(config-router)#network 3.0.0.0 0.255.255.255 area 10
Router-C(config-router)#end
Router-C#
%SYS-5-CONFIG_I: Configured from console by console

Router-C#show ip protocols

Routing Protocol is "ospf 50"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 3.0.0.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    2.0.0.0 0.255.255.255 area 10
    3.0.0.0 0.255.255.255 area 10
  Routing Information Sources:
    Gateway         Distance      Last Update
    3.0.0.1          110           00:00:08
    4.0.0.2          110           00:03:03
    200.100.50.1     110           00:00:33
  Distance: (default is 110)

Router-C#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

O    1.0.0.0/8 [110/200] via 2.0.0.1, 00:09:01, Serial0/3/1
O    2.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C    2.0.0.0/8 is directly connected, Serial0/3/1
C    2.0.0.1/32 is directly connected, Serial0/3/1
L    2.0.0.2/32 is directly connected, Serial0/3/1
O    3.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C    3.0.0.0/8 is directly connected, Serial0/3/0
L    3.0.0.1/32 is directly connected, Serial0/3/0
C    3.0.0.2/32 is directly connected, Serial0/3/0
O    4.0.0.0/8 [110/200] via 3.0.0.2, 00:06:25, Serial0/3/0
```

Router D:

```
Router-D#show ip int brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	unassigned	YES	unset	administratively down	down
FastEthernet0/1	unassigned	YES	unset	administratively down	down
Serial0/3/0	unassigned	YES	unset	administratively down	down
Serial0/3/1	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

```
Router-D#
```



```
Router-D>en
Router-D#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router-D(config)#int serial 0/3/0
Router-D(config-if)#ip address 3.0.0.2 255.0.0.0
Router-D(config-if)#encapsulation ppp
Router-D(config-if)#bandwidth 1000
Router-D(config-if)#clock rate 64000
This command applies only to DCE interfaces
Router-D(config-if)#no shut down

Router-D(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

Router-D(config-if)#end
Router-D#
%SYS-5-CONFIG_I: Configured from console by console

Router-D#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router-D(config)#int serial 0/3/1
Router-D(config-if)#ip address 4.0.0.1 255.0.0.0
Router-D(config-if)#encapsulation ppp
Router-D(config-if)#bandwidth 1000
Router-D(config-if)#clock rate 64000
Router-D(config-if)#no shut down

Router-D(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed state to up

Router-D(config-if)#end
Router-D#
%SYS-5-CONFIG_I: Configured from console by console


Router-D#show ip int brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          200.100.100.1   YES manual up          up
FastEthernet0/1          unassigned      YES unset   administratively down down
Serial0/3/0               3.0.0.2         YES manual up          up
Serial0/3/1               4.0.0.1         YES manual up          up
Vlan1                    unassigned      YES unset   administratively down down
Router-D#


Router-D(config)#int f0/0
Router-D(config-if)#ip address 200.100.100.1 255.255.255.0
Router-D(config-if)#no shut down

Router-D(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router-D(config-if)#end
Router-D#
%SYS-5-CONFIG_I: Configured from console by console
```



```
Router-D#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

```

      3.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       3.0.0.0/8 is directly connected, Serial0/3/0
C       3.0.0.1/32 is directly connected, Serial0/3/0
L       3.0.0.2/32 is directly connected, Serial0/3/0
      4.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C       4.0.0.0/8 is directly connected, Serial0/3/1
L       4.0.0.1/32 is directly connected, Serial0/3/1
C       4.0.0.2/32 is directly connected, Serial0/3/1
      200.100.100.0/24 is variably subnetted, 2 subnets, 2 masks
C       200.100.100.0/24 is directly connected, FastEthernet0/0
L       200.100.100.1/32 is directly connected, FastEthernet0/0
```

```
Router-D(config)#router ospf 50
Router-D(config-router)#network 3.0.0.0 0.255.255.255 area 10
Router-D(config-router)#network
00:48:44: %OSPF-5-ADJCHG: Process 50, Nbr 3.0.0.1 on Serial0/3/0 from LOADING to FULL, Loading
Done
```

% Incomplete command.

```
Router-D(config-router)#network 4.0.0.0 0.255.255.255 area 10
Router-D(config-router)#
00:49:16: %OSPF-5-ADJCHG: Process 50, Nbr 4.0.0.2 on Serial0/3/1 from LOADING to FULL, Loading
Done
```

```
Router-D(config-router)#end
Router-D#
%SYS-5-CONFIG_I: Configured from console by console
```

```
Router-D#show ip protocols
```

```
Routing Protocol is "ospf 50"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 200.100.100.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    3.0.0.0 0.255.255.255 area 10
    4.0.0.0 0.255.255.255 area 10
  Routing Information Sources:
    Gateway         Distance      Last Update
    3.0.0.1          110          00:00:39
    4.0.0.2          110          00:00:07
    200.100.50.1     110          00:02:44
    200.100.100.1    110          00:00:07
  Distance: (default is 110)
```

```
Router-D#show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP  
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP  
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area  
* - candidate default, U - per-user static route, o - ODR  
P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
O    1.0.0.0/8 [110/200] via 4.0.0.2, 00:03:42, Serial0/3/1  
O    2.0.0.0/8 [110/200] via 3.0.0.1, 00:04:14, Serial0/3/0  
    3.0.0.0/8 is variably subnetted, 3 subnets, 2 masks  
C    3.0.0.0/8 is directly connected, Serial0/3/0  
C    3.0.0.1/32 is directly connected, Serial0/3/0  
L    3.0.0.2/32 is directly connected, Serial0/3/0  
    4.0.0.0/8 is variably subnetted, 3 subnets, 2 masks  
C    4.0.0.0/8 is directly connected, Serial0/3/1  
L    4.0.0.1/32 is directly connected, Serial0/3/1  
C    4.0.0.2/32 is directly connected, Serial0/3/1  
    200.100.100.0/24 is variably subnetted, 2 subnets, 2 masks  
C    200.100.100.0/24 is directly connected, FastEthernet0/0  
L    200.100.100.1/32 is directly connected, FastEthernet0/0
```

Ping Test:

```
Cisco Packet Tracer PC Command Line 1.0  
C:\>ping 200.100.100.2  
  
Pinging 200.100.100.2 with 32 bytes of data:  
  
Reply from 200.100.50.1: Destination host unreachable.  
Reply from 200.100.50.1: Destination host unreachable.  
Reply from 200.100.50.1: Destination host unreachable.  
Reply from 200.100.50.1: Destination host unreachable.  
  
Ping statistics for 200.100.100.2:  
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),  
  
C:\>ping 200.100.100.2  
  
Pinging 200.100.100.2 with 32 bytes of data:  
  
Reply from 200.100.50.1: Destination host unreachable.  
Reply from 200.100.50.1: Destination host unreachable.  
Reply from 200.100.50.1: Destination host unreachable.  
Reply from 200.100.50.1: Destination host unreachable.  
  
Ping statistics for 200.100.100.2:  
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),  
  
C:\>ping 200.100.100.2  
  
Pinging 200.100.100.2 with 32 bytes of data:  
  
Reply from 200.100.50.1: Destination host unreachable.  
Reply from 200.100.50.1: Destination host unreachable.  
Reply from 200.100.50.1: Destination host unreachable.  
Reply from 200.100.50.1: Destination host unreachable.  
  
Ping statistics for 200.100.100.2:  
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```